RUHR-UNIVERSITÄT BOCHUM

EXTERNAL STORAGE

Computer Architecture

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Agenda

- Hard drive disks
- Solid state disks
- Virtual Memory
- Magnetic Tapes

Hard Disks

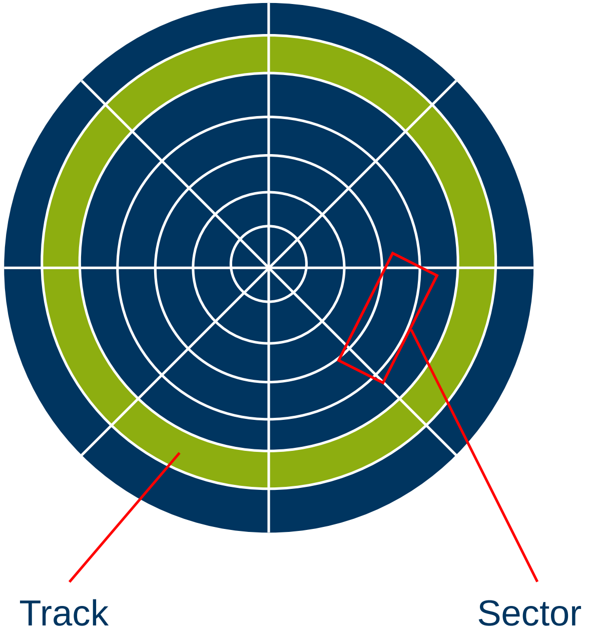
Hard Drive Disk: Disk Layout

Memory Disk where data is stored on moving platters

Hard Drive Disk: Disk Layout

Memory Disk where data is stored on moving platters

- 4-8 platters per Disk
- Platters are divided into tracks and sectors
- Sector size: 128 Byte – 1 Kilobyte
- Platter rotation speed: up to 10,800 rpm (revolutions per minute)



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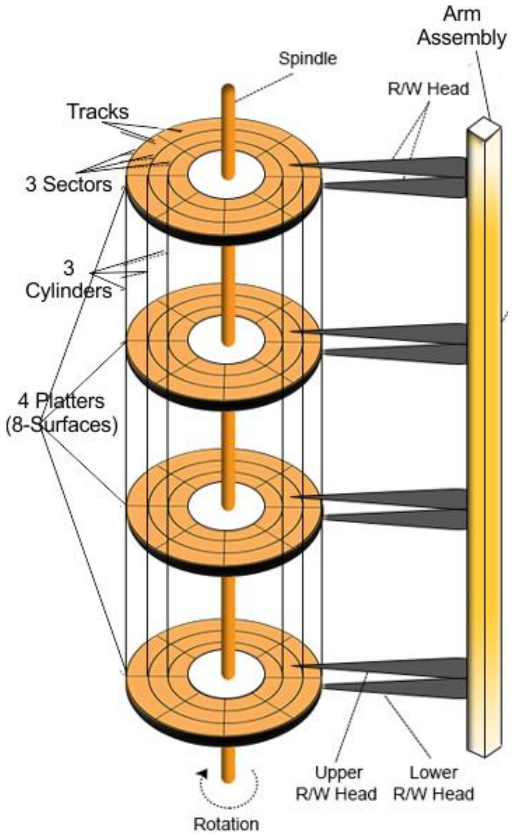
HDD: Physical View

All platters of a disk are put on a spindle with read-write heads in between

RW-heads are fixed in position and can only move towards or from the spindle to access different tracks

Each platter stores data on both sides



HDD: Physical View

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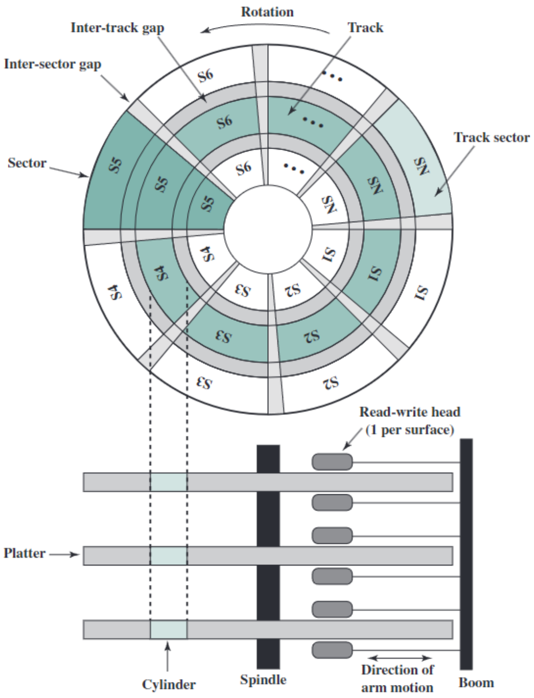
To access rotate platter and put RW-head in position to access a sector

Because of moving parts HDD's can scratch easily when they are dropped or moved while in use



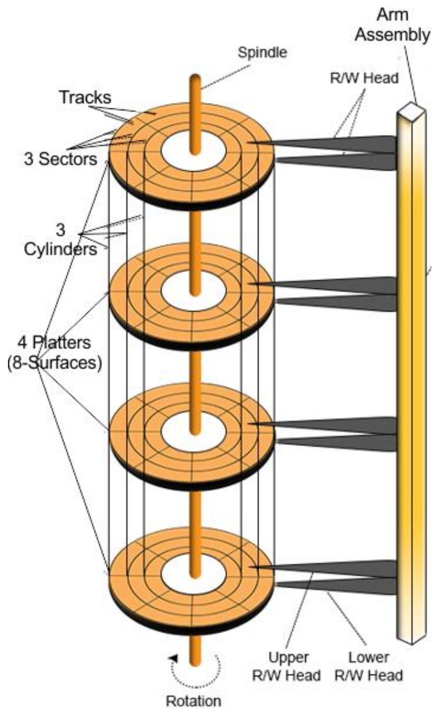
"Disc Data Layout" from "Computer Organization and Architecture" by William Stallings 11. Edition, page 235

How Data Is Addressed: CHS versus LBA

How do you find the data you need on a HDD?

CHS-Coordinates:

- C: Cylinder (Track on a disc)
- H: Head (Disc used)
- S: Sector



Early disks did not come with a disk controller: OS did all accesses with help of geometric disk overview

Physical location of data is exposed to computer

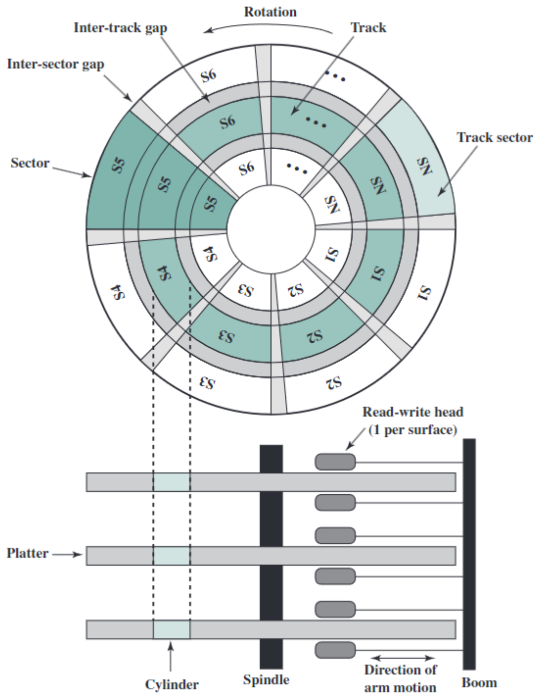
CHS versus LBA

Logical Block Addressing:

- Each sector is given a consecutive integer 0…n

The translation of physical blocks to virtual addresses is done by a disk controller

Physical address of data is not exposed



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Partitioning: What and Why?

Divide the memory of e.g. a HDD into multiple sections

Why?

- Install multiple OS
- Enhance data security: if there is an error in one partition the others are not affected
- Organize data: use multiple data systems, make backups easier
- Make system more secure: set different properties for each partition e.g. read-only

Partitioning: How?

BIOS-based systems: Master Boot Record (MBR)

- Consists of bootstrap code and a partition table that keeps track of all partitions

UEFI-based systems: GUID-Partition table (GPT)

- **G**lobally **U**nique **Id**entifier per partition
- Consists of MBR, primary and secondary partition table
- Redundancy helps with recovering the table if an error occurs

HDD Access Times

T_h: Move head to correct cylinder

T_s : Wait until sector rotates to head (∅ Time: 0.5 rotations)

T_t : Transfer sector contents

$$T_{total} = T_h + T_s$$

Why isn't T_t part of the formula?

T_h and T_s are measured in msec while T_t is measured in microseconds

→ can be neglected

And now it's your turn…

An HDD has the following specifications:
Its platters move with a speed of 3600 rpm and its head positions at the right track within 10msec.
What is the access time T_{total} in milliseconds (msec)?

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Its platters move with a speed of 3600 rpm and its head positions at the right track within 10msec.
What is the access time T_{total} in milliseconds (msec)?

Disk speed = 3600 rpm = 60 rps
Time for one full rotation = 1/60s = 16.67ms
Average rotational latency = 8.33 ms

T_{total} = 10 + 8.33 ms = 18.33 ms

SSD/Flash

Solid State Drives: Physical Composition

No moving parts

Instead Flash-Memory (energy efficient) or Synchronous DRAM (fast)

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A Flash Cell

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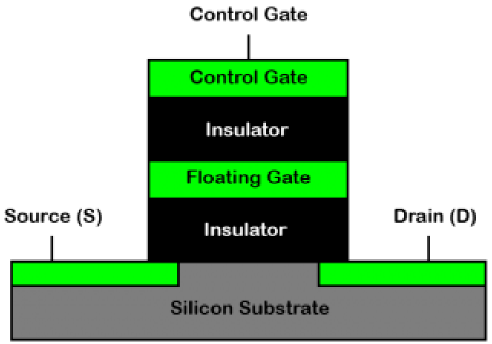
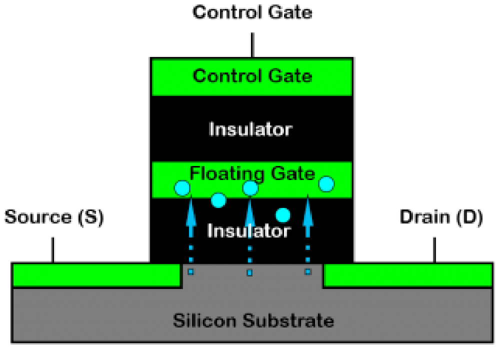


A Flash Cell

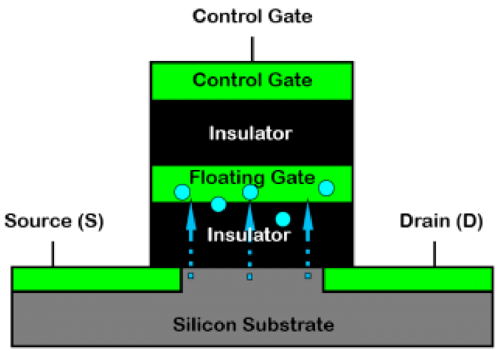
Solid State Drives: Physical Composition



Storing data

Credits: https://www.syslogic.com/blog/how-ssd-nand-flash-storage-works

Solid State Drives: Physical Composition



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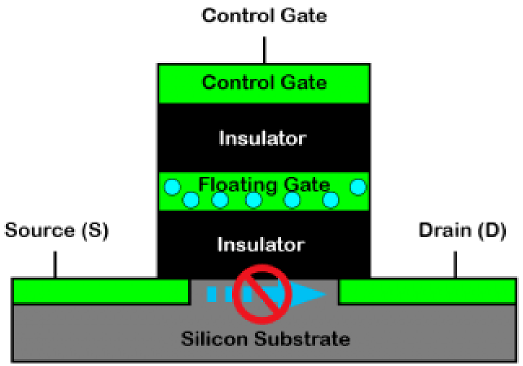

Reading data

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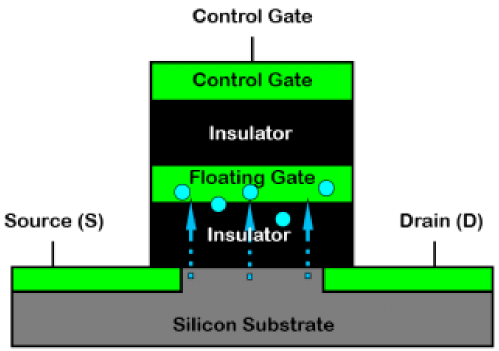
Solid State Drives: Physical Composition



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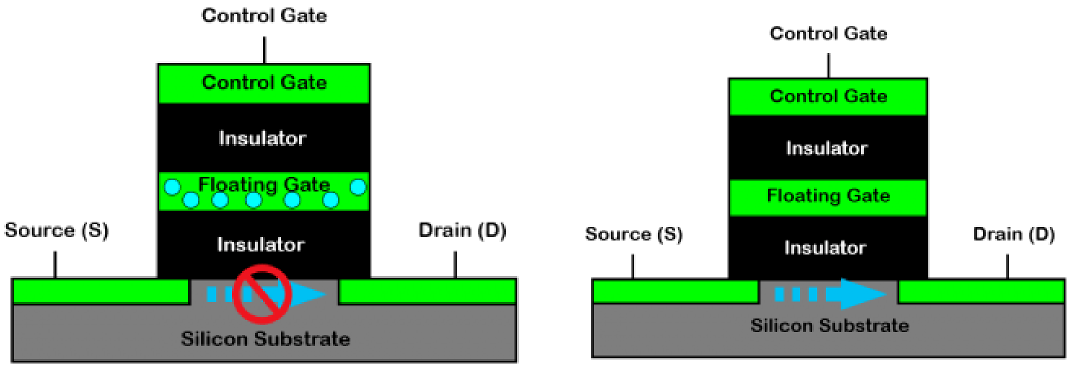

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SSD versus HDD

SSD:

- **No moving parts: no wear and resistance to movement and dust**
- **Better for devices where minimal power use is necessary**
- **Expensive**
- **Flash memory can only be rewritten a finite number of times (replacement cells are put into disk)**

HDD:

- **Moving parts: wear possible, has to be handled carefully**
- **Needs more energy**
- **Cheaper**
- **Can be rewritten an infinite amount of times**

Virtual Memory

Most Computers today are capable to access more memory than can be realistically installed

If bus width is n, 2^n memory cells can be addressed

Bus width	Addressable memory cells	Cost if 100€/16GB
16	2^16	--
32	2^32	ca. 25 €
64	2^64	ca. 105 Billion €

→ it gets unrealistic…

Virtual Memory

Idea: use disks to virtually enlarge main memory

How to combine main memory and disks?

- **Segmentation**
- **Paging**

Every program comes with a virtual address space

Segmentation

Divide virtual address space of a program into segments

OS can decide which segments to keep in main memory and which to move to a disk

Is not used anymore today

Paging

Divide virtual memory into pages of fixed size and put all of them onto a disc

Main memory consists of "page frames" that can take one page each

A "page table" keeps track of which page frame is used by which page

Main Memory
Page Frame 0
Page Frame 1
Page Frame 2
Page Frame 3

Valid	Disk Address	Page
1	0011101010010101111	001
1	0011101010010100001	110
0		
0		

Virtual Memory
Page 0
Page 1
Page 2
Page 3
Page 4
Page 5
Page 6
Page 7

Paging

Usage similar to caches:

Check if page is currently in main memory

Page hit: use this entry

Page miss: check if there is an unused page frame, if not replace one determined by a replacement strategy

Use unused page frame to load needed frame into main memory

Tapes

Magnetic Tapes

PVC tape coated in a magnetizable material

Advantages:

- **high capacity (today: more than 18 Tb per tape)**
- **good sequential access time**
- **long lasting (10-30 years)**
- **can be rewritten multiple times**

Disadvantages:

- **sensitive to magnetic fields and dust**
- **high random access time (60 seconds)**
- **data can only be added at the end**

Who Still Uses Tape?

Use case today: archive loads of data

Global Memory need grows by 30-40% each year: disks cannot keep up

A lot of data does not need to be accessed instantly e.g. scientific archives, backups

Tape is also reliable and does not consume power when inactive

Even Google and Microsoft use robotic tape libraries

Sun T7000 Tape Library Robot

Summary

- HDDs are a type of disk where data is put onto moving platters that rotate and are accessed through a read-write head
- SSDs have no moving parts and instead use Flash Memory (or SDRAM)
- HDDs are cheap but sensitive to dust and movement while SSDs are less sensitive to outside influences but more expensive
- A program’s address space can be extended by using a virtual address space and storing some of the data on disks
- Magnetic tapes are inexpensive. They are still used for archives or backups. Slow random access, good sequential access times.